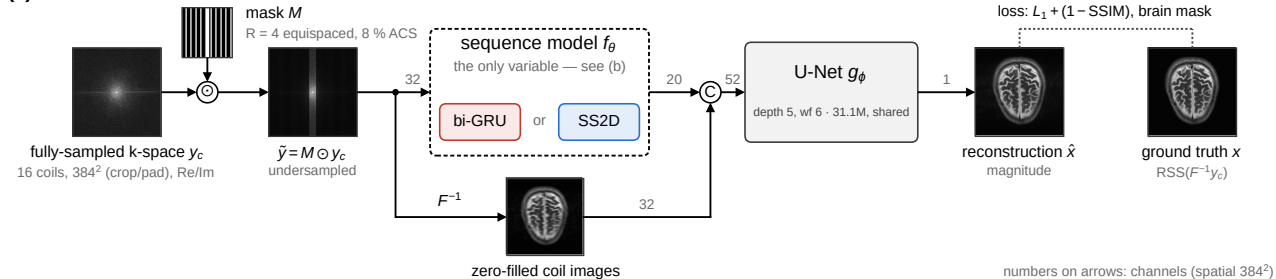
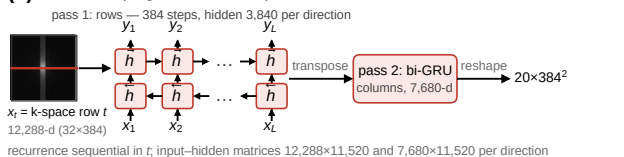


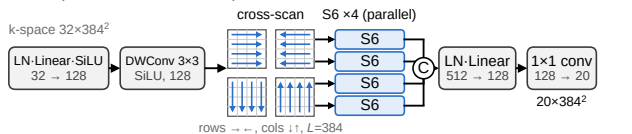
(a) shared ETER-Net pipeline



(b) bi-GRU (original ETER-Net)



SS2D (controlled substitution)



$$S6: h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t + D x_t, \quad (\Delta_t, B_t, C_t) \text{ from } x_t$$

d_inner 128, d_state 16; one S6 weight set per direction, shared by all its rows (columns)

34.2M (SSM stack 3.1M), fp16 scan

(c) enhanced SS2D (replaces f_θ in (a))

